Lab 5. Brushless direct current motor control

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Specialization: Automation

Objective

Introduction to methods of flux-weakening control for permanent magnet synchronous motors.

Initial data

Parameter	R_s	L_s	Ψ_f	Z_p	J	U_{DC}	T_S
Value	**14.7282**	**0.0311**	**0.1250**	**16**	**4.7639**	**48**	**0.0001**

1. Assemble a mathematical model of a BLDC motor in Matlab/Simulink.

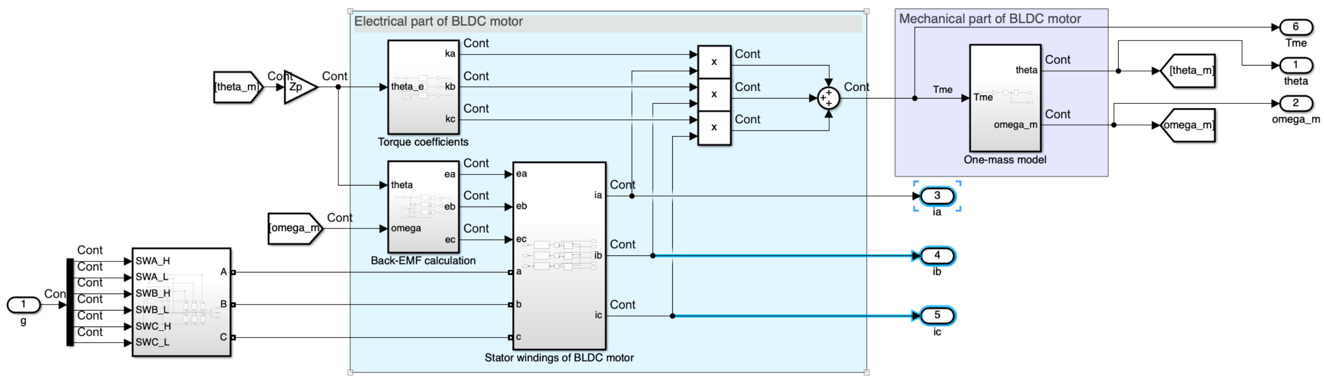

Figure 1. Model of BLDC drive

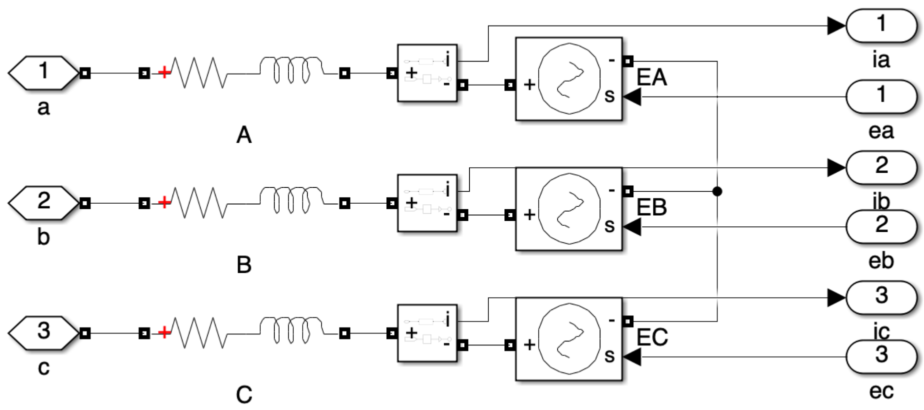

Fig2. Model of stator windings of BLDC motor

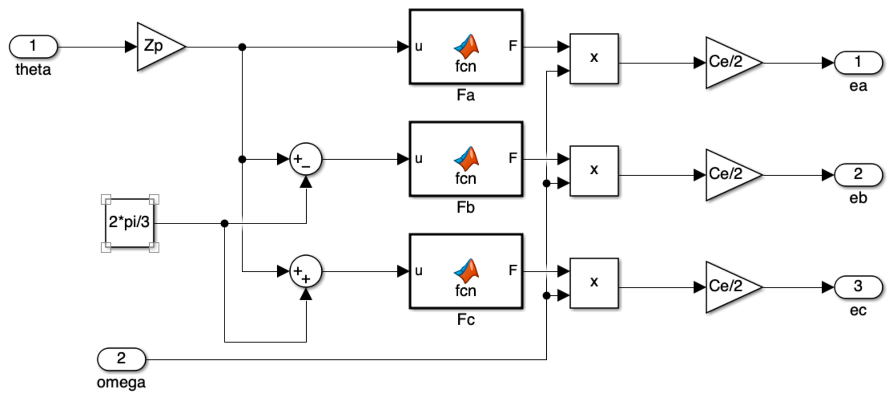

Figure 3. Model of back-EMF calculation for BLDC motor

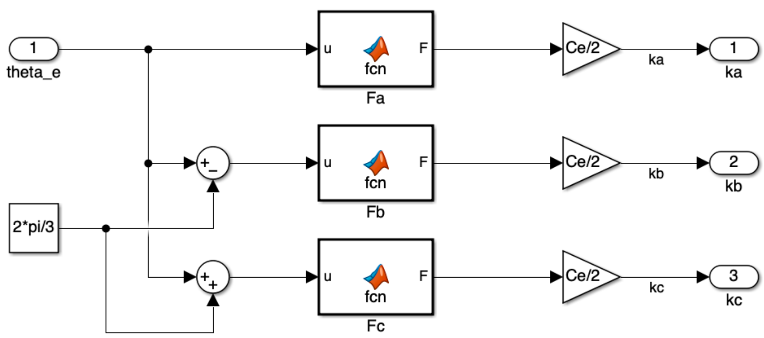

Figure 4. Model of torque constants calculation for BLDC motor

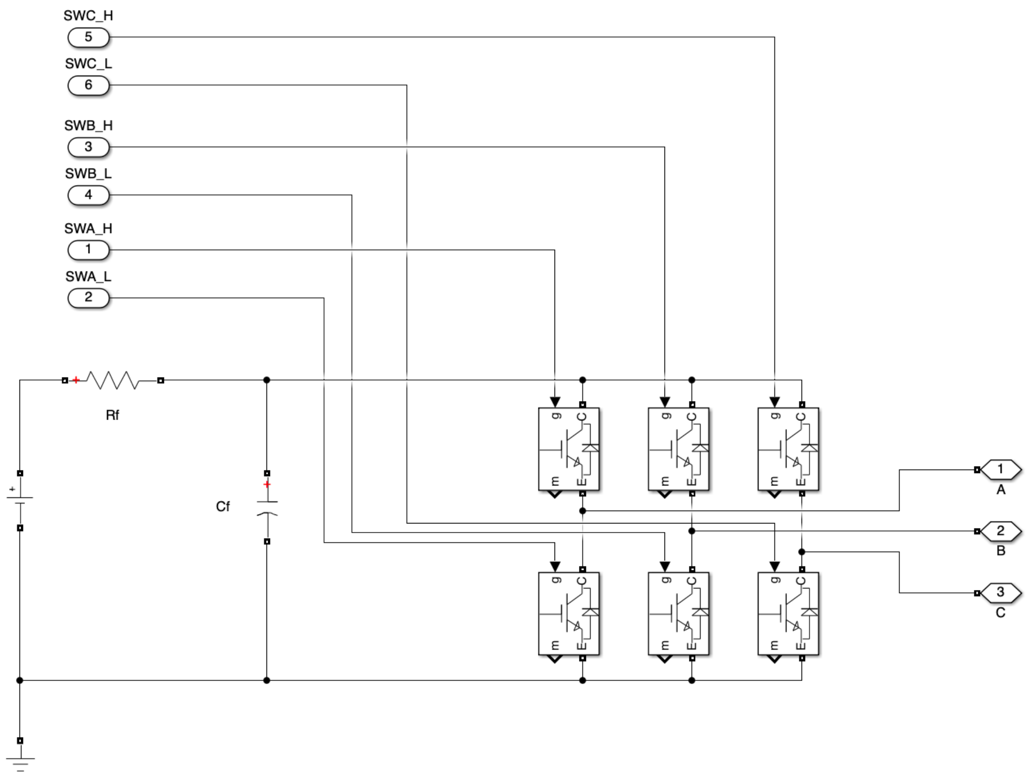

Figure 5. Model of three-phase inverter for BLDC motor

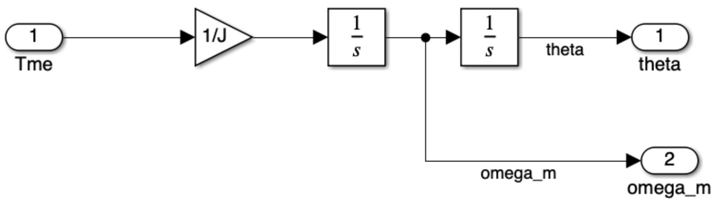

Figure 6. Mechanical part of BLDC motor

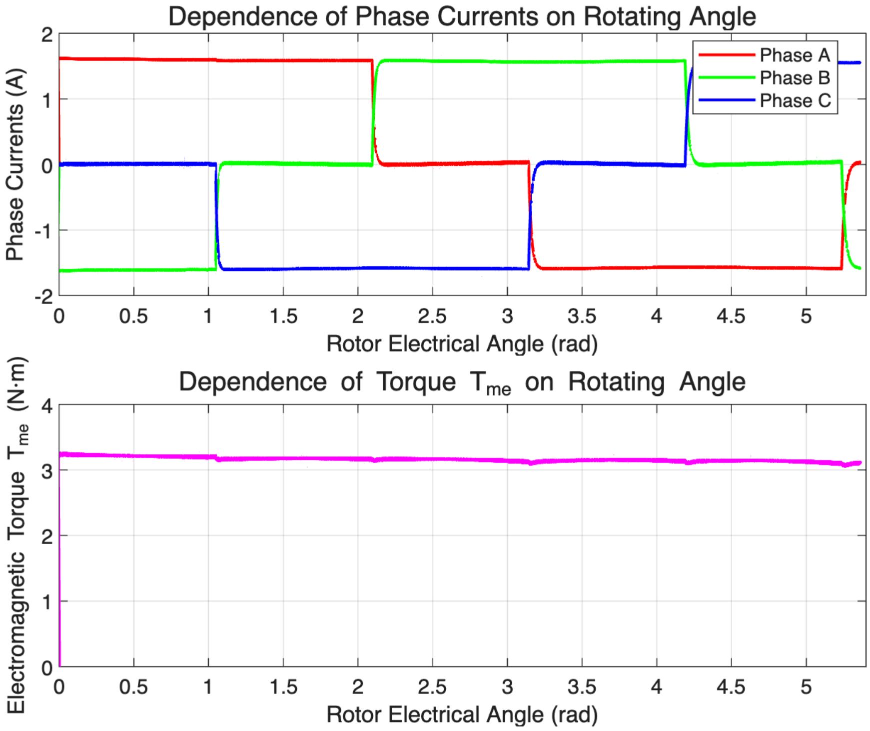

Fig7. dependences of phase currents and torque T_{me} on rotating angle.

2. Tune up current loop.

Linear optimum transfer function of open loop system:

$$W_{\text{et.oploop}}(s) = \frac{1}{T_T s + 1}$$

Calculated values for PI coefficients:

$K_p = 0.3068$

$K_i = 145.3$

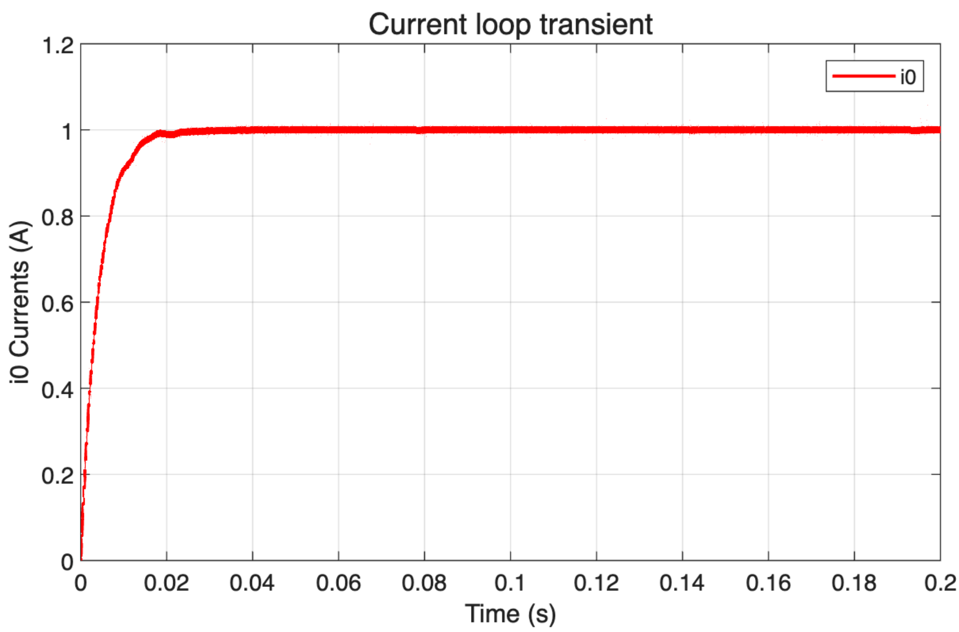

Figure 8. Current loop transient.

Transient time = 0.015

Overshoot = 5%

3. Tune up speed loop.

Symmetrical optimum transfer function of open loop system:

$$W_{\text{et.oploop}}(s) = \frac{4T_u s + 1}{8T_u{}^2 s^2 (T_u s + 1)}$$

Small speed $\omega_{\text{ref}} = 0.001\omega_{\max}$.

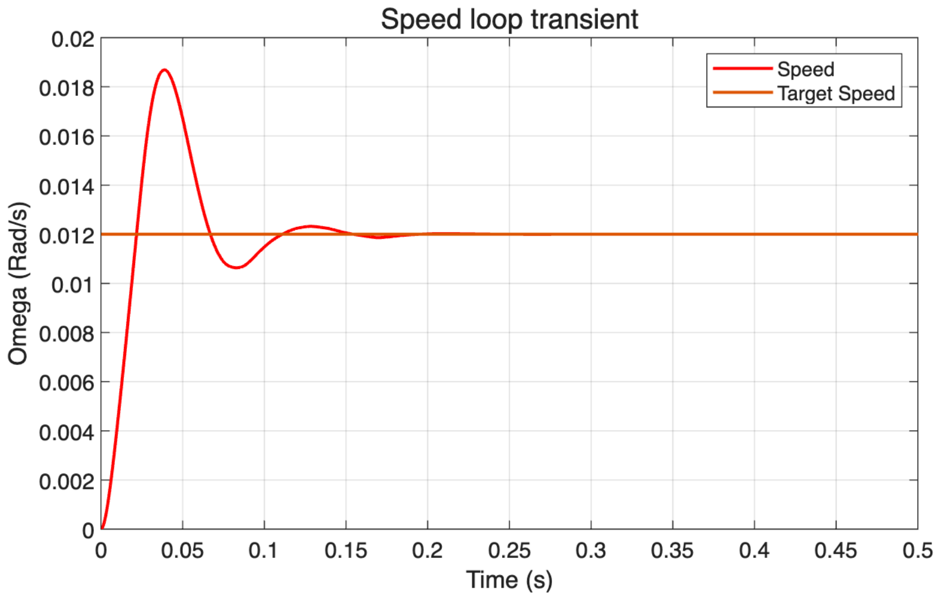

Figure 9. Speed loop transients.

Transient time = 0.1 s

Overshoot = 55%

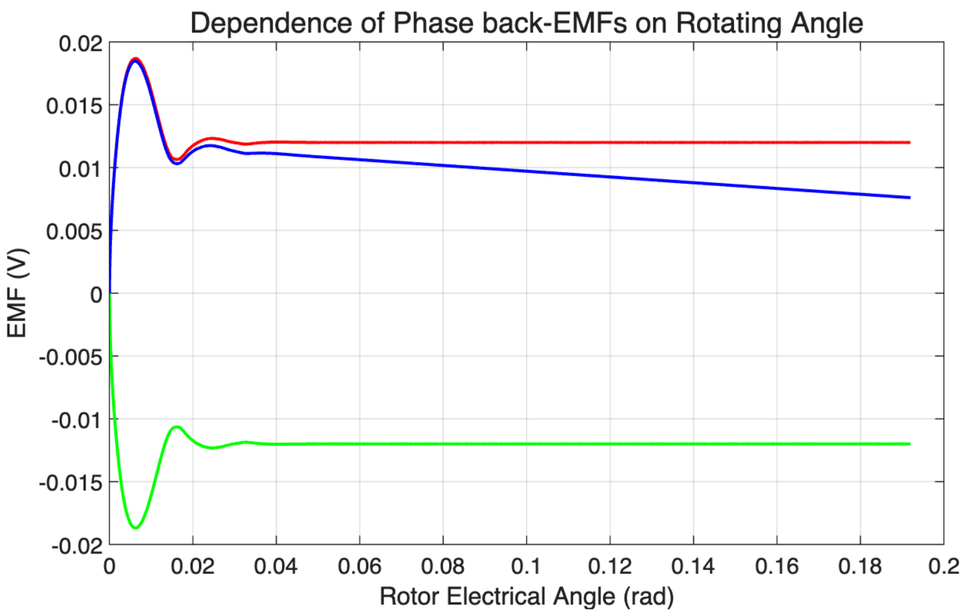

Figure 5. Dependences of phase back-EMFs on rotating angle.

4. Simulation with the presence of disturbance torque.

Add disturbance Tload = 1 at t=0.3s:

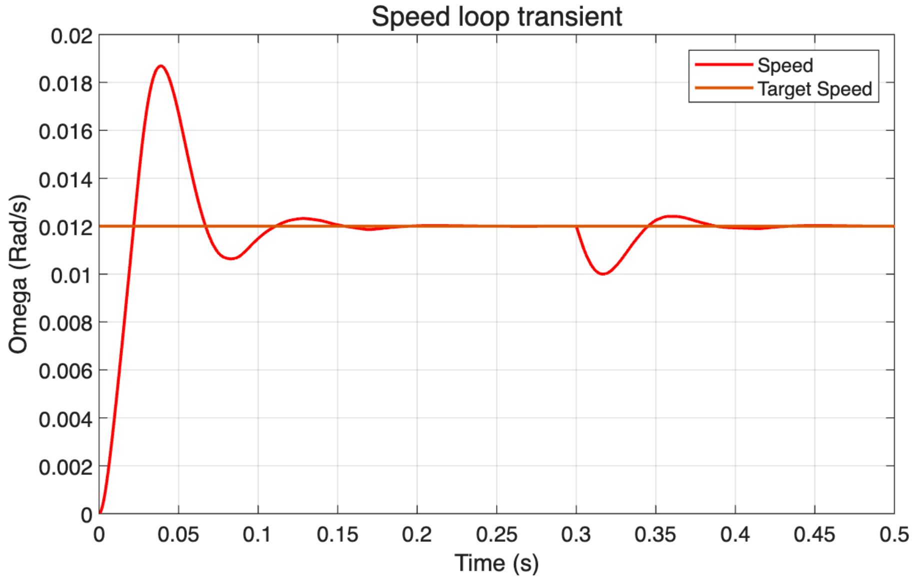

Figure 10. Speed loop transients.

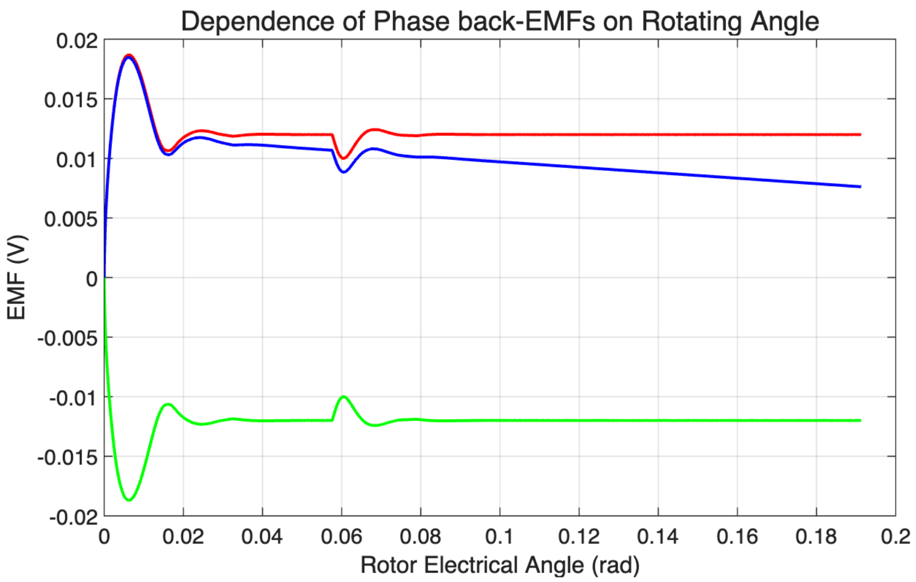

Figure 7. Dependences of phase back-EMFs on rotating angle

Conclusions:

A complete mathematical model of the BLDC motor drive was implemented and verified. The simulation correctly produced the characteristic trapezoidal back-EMF waveforms, phase currents, and torque profiles. The current loop, tuned using the Linear Optimum method, demonstrated excellent performance with a fast transient time (0.015 s) and low overshoot (5%), ensuring precise torque control.

Disturbance Rejection Capability: The system's robustness was tested by applying a step disturbance torque. The simulation showed that the tuned speed controller was able to recover and regulate the speed back to its reference value after the disturbance. This confirms that the control system, designed with the symmetrical optimum, effectively fulfills its primary purpose of strong disturbance rejection, maintaining stable operation under load variations.

The speed controller, tuned via the Symmetrical Optimum method, effectively regulates speed and rejects load disturbances. As expected with this method, the response to a reference change exhibits significant overshoot (55%) but provides strong recovery from a step disturbance torque, confirming a high degree of robustness.